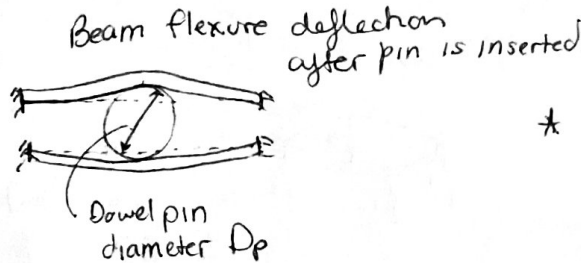
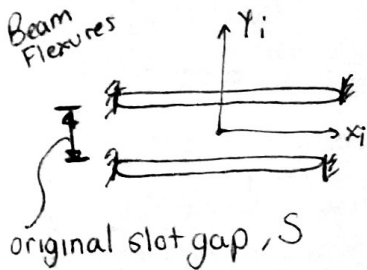


EAC Model Derivation

Rachel Hoffman

Based on the model described in "Principle of elastic averaging for rapid precision design"

by Teo and Slocum.

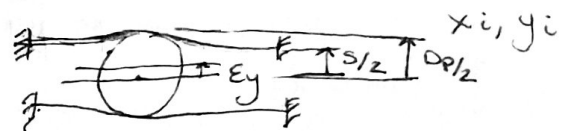


* Manufacturing error in flexures and/or improper placement of dowel pin will cause a net displacement in the local coordinate system

The total deflections in the beams are

$$(I) \quad \sigma_{Top} = \left(\frac{D_p}{2} - \frac{S}{2} \right) + \epsilon_y$$

$$(II) \quad \sigma_{Bottom} = \left(\frac{D_p}{2} - \frac{S}{2} \right) - \epsilon_y$$



For a fixed-fixed beam, the deflection due to a load at the midpoint is

$$\sigma = \frac{F l^3}{192 EI}$$

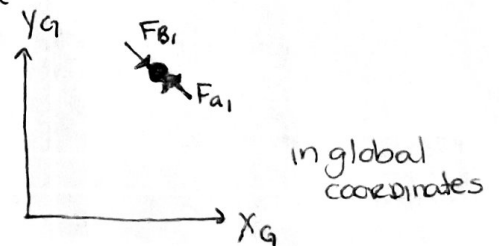
Stiffness is defined as $\frac{F}{\sigma}$, so stiffness $K = \frac{192 EI}{l^3}$

so the net force acting on the dowel pins are

$$(III) \quad F_{ai} = \left(\frac{192 EI}{l^3} \right) \left(\frac{D_p}{2} - \frac{S}{2} + \epsilon_y \right)$$

$$(IV) \quad F_{bi} = \left(\frac{192 EI}{l^3} \right) \left(\frac{D_p}{2} - \frac{S}{2} - \epsilon_y \right)$$

$$F_{net i} = F_{ai} - F_{bi}$$



looking at springs connected by a rigid link (the plate in this case) the deflection of one point is coupled to the deflection at another.

I find it easiest to view as a ring with springs attached

l_{si} = length of spring i

→ deflection of spring 1, ϵ_1

Deflection of spring 2, ϵ_2



$$l_{s1} = l_{s2} = l_{s3} = l_{s4}$$



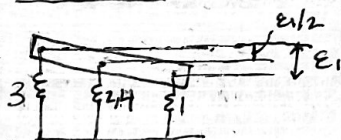
$$l_{s1} = (l_{s10} - \epsilon_1)$$

$$l_{s2} = l_{s20} - \frac{1}{2}\epsilon_1$$

$$l_{s3} = l_{s30}$$

$$l_{s4} = l_{s40} - \frac{1}{2}\epsilon_1$$

SIDE VIEW



A deflection of spring 1 by ϵ_1 results in a coupled deflection of spring 2 and 3 by $\frac{\epsilon_1}{2}$



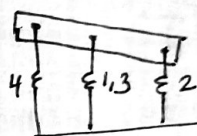
$$l_{s1} = l_{s10} - \frac{\epsilon_2}{2}$$

$$l_{s2} = l_{s20} - \epsilon_2$$

$$l_{s3} = l_{s30} - \frac{\epsilon_2}{2}$$

$$l_{s4} = l_{s40}$$

FRONT VIEW



A deflection of spring 2 by ϵ_2 results in a coupled deflection of spring 1 and 3 by $\frac{\epsilon_2}{2}$

We can find similar relations for spring 3 deflected by ϵ_3 and spring 4 deflected by ϵ_4

- what if all these unique deflections ($\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4$) happened simultaneously? what would the net displacement of each spring be?

$$\delta_1 = \epsilon_1 + \frac{\epsilon_2}{2} + 0 + \frac{\epsilon_4}{2}$$

$$\delta_2 = \frac{\epsilon_1}{2} + \epsilon_2 + \frac{\epsilon_3}{2} + 0$$

$$\delta_3 = 0 + \frac{\epsilon_2}{2} + \epsilon_3 + \frac{\epsilon_4}{2}$$

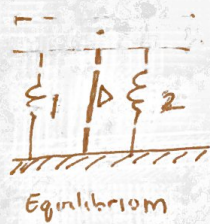
$$\delta_4 = \frac{\epsilon_1}{2} + 0 + \frac{\epsilon_3}{2} + \epsilon_4$$

$$\begin{array}{cccc} \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ \text{due to } \epsilon_1 & \text{due to } \epsilon_2 & \text{due to } \epsilon_3 & \text{due to } \epsilon_4 \end{array}$$

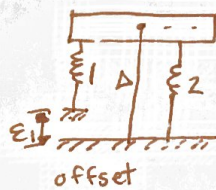
These are the resultant displacements due to offsets in the pin position (ϵ_y in eqn I-IV)

is this the final displacement? Not exactly. These offsets cause a "ripple effect" of force balances to reach an equilibrium position.

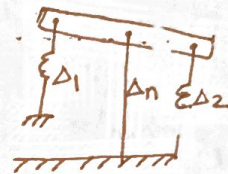
For example, say we have two springs and we force the displacement of one of the springs.



$\Rightarrow$



$\Rightarrow$

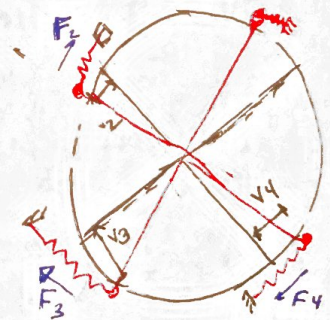
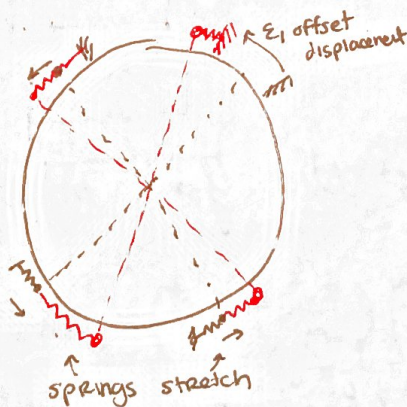
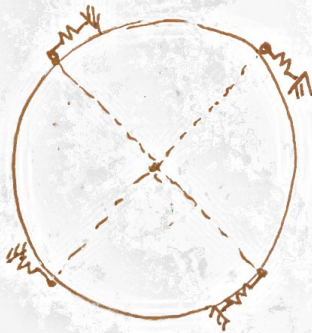


$$\Delta_n = \frac{K\Delta_1 + K\Delta_2}{2}$$

$$\Delta_n = K\left(\frac{\Delta_1 + \Delta_2}{2}\right)$$

basically, the equilibrium position is the average of the spring positions

lets look at an example with just one offset



$F =$ "stretch reaction forces"

These reaction forces cause reaction displacements that bring the system to an equilibrium position

reaction displacements

V_1, V_2, V_3, V_4

* The reaction displacements follow the same spring-rigid body constraints as the offset displacements.

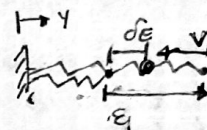
We can predict the equilibrium position of each of the springs by dividing the resultant displacements due to the offset of the pin positions by the number of spring elements.

(VI)

$$\sigma_{1E} = \sigma_1 / 4, \quad \sigma_{2E} = \frac{\sigma_2}{4}, \quad \sigma_{3E} = \frac{\sigma_3}{4}, \quad \sigma_{4E} = \frac{\sigma_4}{4}$$

Approximated equilibrium positions

Then, the reaction displacements V_1, V_2, V_3, V_4



equilibrium position

you return to position in the 'negative dir'

Reaction displacements

The balanced displacements = $d_{1E}, d_{2E}, d_{3E}, d_{4E}$

The displacement vectors are

$$V_1 = \epsilon_1 - \sigma_{1E}$$

$$V_2 = \epsilon_2 - \sigma_{2E}$$

$$V_3 = \epsilon_3 - \sigma_{3E}$$

$$V_4 = \epsilon_4 - \sigma_{4E}$$

(VII)

now we can find the forces in the system

(VIII)

$$F_{aid} = \left(\frac{192 EI}{l^3} \right) \left(\frac{PP}{2} - \frac{S}{2} + \sigma_i \right)$$

$$F_{bid} = \left(\frac{192 EI}{l^3} \right) \left(\frac{PP}{2} - \frac{S}{2} - \sigma_i \right)$$

where σ_i is the displacement of each flexure pair due to the offsets, $\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4$

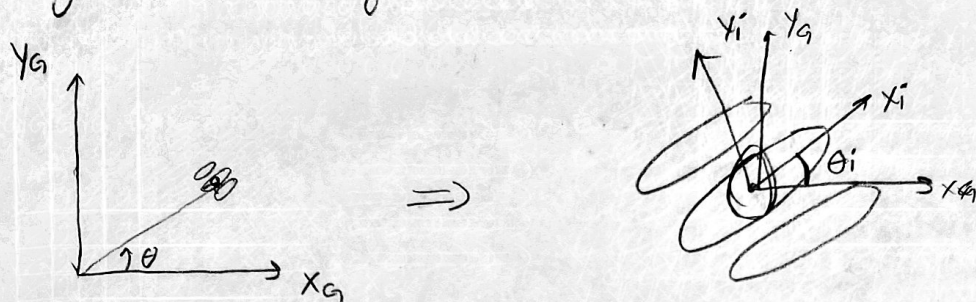
(IX)

$$F_{air} = \left(\frac{192 EI}{l^3} \right) \left(\frac{PP}{2} - \frac{S}{2} + V_i \right)$$

$$F_{bir} = \left(\frac{192 EI}{l^3} \right) \left(\frac{PP}{2} - \frac{S}{2} - V_i \right)$$

where V_i is the displacement vectors driving the system to equilibrium from (VII)

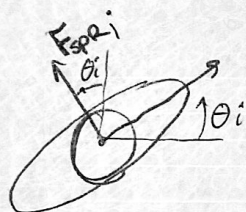
Finally, we can find the net forces acting on the system in the global coordinates



$$X_g = X_i \cos \theta_i - Y_i \sin \theta_i$$

$$Y_g = Y_i \cos \theta_i + X_i \sin \theta_i$$

Apply this transformation to forces



$$F_x = -F_{spri} \sin \theta_i \quad (\text{X})$$

$$F_y = F_{spri} \cos \theta_i$$

where $F_{spri} = (F_a - F_b)$ (XI)



Combining (VIII, IX, X and XI) we can find the net forces acting on the entire system.

$$(XII) F_x = \sum -((F_{aid}) - (F_{bid})) \sin \theta_i + \sum -((F_{air}) - (F_{bir})) \sin \theta_i$$

$$(XIII) F_y = \sum ((F_{aid}) - (F_{bid})) \cos(\theta_i) + \sum ((F_{air}) - (F_{bir})) \cos \theta_i \quad i = 1, 2, 3, 4$$

Or # flexure features if expanded)

$$(XIV) M_z = \sum ((F_{aid} - F_{bid}) \cdot r_i) + \sum ((F_{air} - F_{bir}) \cdot r_i)$$

$$(XV) \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta \theta_z \end{bmatrix} = \begin{bmatrix} F_x / k_x \\ F_y / k_y \\ M_z / k_{\theta_z} \end{bmatrix}$$

what are k_x, k_y, k_{θ} ?

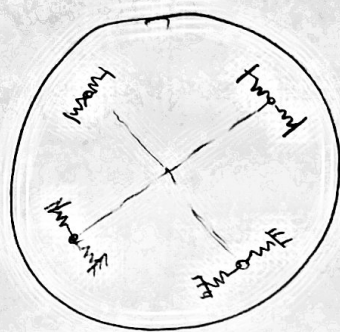
For x and y

Forces and displacements

all 8 springs act in parallel so their

stiffnesses add together.

They all push or pull to bring the system back to equilibrium.



one spring for each beam flexure.

$$K_x = (8) \left(\frac{192EI}{e^3} \right) \sin(\theta)$$

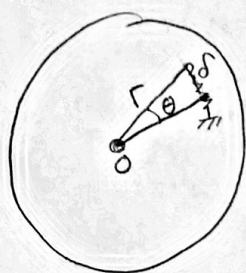
$$K_y = (8) \left(\frac{192EI}{e^3} \right) \cos(\theta)$$



For the rotational stiffness, K_θ we have to take two things into account.

The first is that the springs are no longer acting in parallel, they are acting in series and only four of them at a time will provide stiffness about either $+\theta$ or $-\theta$.

$$K_{eq} = \frac{1}{\left(\frac{1}{k}\right) + \left(\frac{1}{k}\right) + \left(\frac{1}{k}\right) + \left(\frac{1}{k}\right)} = \frac{1}{4\left(\frac{1}{k}\right)} = \frac{k}{4} = K_{eq}$$



$$F_{spr} = K_{eq} \delta \quad \delta = r\theta \quad (\text{small } \theta \text{ approx})$$

$$F_{spr} = K_{eq} r \theta$$

$$\text{Moment about } O \quad M = F_{spr} \cdot r$$

$$M = (K_{eq} r \theta) \cdot r = K_{eq} \theta r^2 = M$$

$$\theta = \frac{M}{K_{eq} r^2} \rightarrow K_\theta = \frac{M}{\theta} = K_{eq} r^2 = K_\theta \quad (\text{XVII})$$

* note, the system stiffness described by Teo and Slocum is a bit more complicated as it involves jacobians and rotation matrices as well as beam stiffness matrices. My way is slightly more intuitive (to me) but it may result in "rousher" estimations.